

# JOCELYN XU

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## EDUCATION

### University of Waterloo

Computer Science, Honours, Co-op Program

Sept 2024 – Apr 2028

Waterloo, ON

- CAV: 95
- Courses: Data Structure & Algorithms, Object Oriented Programming, Functional Programming, Software Design

## EXPERIENCE

### Bank of Montreal (BMO) Software Developer | *Angular, TypeScript, AWS, Web Components*

May 2025 – Dec 2025

#### Help Center AI Chatbot

- Architected a serverless RAG chatbot using **AWS Lambda** and **Amazon Bedrock**, integrating **Claude 3.7 Sonnet** and **WebSockets** for real-time response streaming, increasing Help Center engagement by **30%**
- Engineered a **multi-agent LLM routing system** using a supervisor-collaborator architecture to classify query intent and select specialized agents, reducing inference latency by **35%** through optimized routing and **LRU caching**
- Eliminated **20+** monthly hours of manual work by building a scheduled **Node.js/Puppeteer ETL pipeline** to process **125+** banking documents into **Amazon S3** for real-time RAG retrieval via Bedrock Knowledge Bases

#### Lexicon Design System (LDS)

- Scaled BMO's design system to **300+ developers** by building **15+** web components with TypeScript, CSS, and Storybook, replacing legacy Angular components and ensuring **WCAG 2.2 AA compliance**
- Improved code coverage to **80%+** by authoring **250+** Jest and Playwright tests, resolving **360+** bugs in shared components, preventing repeated defects and reducing QA effort across teams
- Reduced development timelines by **3+ months** across bi-weekly releases by publishing versioned design system components as internal **NPM packages** via **JBfrog Artifactory**, supporting BMO products serving **60,000+** clients

### Data Science Club VP Development | *Typescript, React, Next.js, PostgreSQL, Shadcn/ui*

Dec 2024 – Apr 2026

- Led a team of **11 developers** to deliver **3 full-stack web apps** serving **5,000+** users by defining the project architecture, designing database schemas, scoping technical milestones, and conducting code reviews
- Spearheaded a full-stack hackathon portal serving **1,100+** users using **TypeScript, React**, and **REST APIs**, implementing a QR code and NFC tag check-in system that reduced manual processing time from **60s** to under **5s**
- Reduced project management and documentation overhead by architecting **GitHub Actions** CI/CD with quality gates, centralized **secret management**, and an **agentic documentation pipeline** based on pull request differences

## PROJECTS

### Catan Solver | *PyTorch, OpenCV, YOLOv8, Multimodal AI Agents, Statistic Models*

Jan 2026

- Engineered a hybrid multi-threaded AI agent capable of playing competitive Catan, integrating computer vision, probabilistic reasoning, and LLM-based decision making
- Fine-tuned a **YOLOv8m-seg** model on **500+** annotated screenshots to detect hex tiles and ports, achieving **98.5%** mAP with sub-**100ms** inference latency
- Achieved a **45%** win rate by designing a Bayesian uncertainty engine tracking opponent hands at **92%** accuracy and Monte Carlo simulations evaluating **10,000+** board states per turn

### Fashionkilla | *React, Tailwind, MongoDB, FastAPI, Python, Sentence Transformers*

Oct 2025

- Recognized with **Best Use of MongoDB** and **Best UI/UX** at TechNova 2025 Hackathon
- Engineered an AI fashion recommendation engine using Sentence Transformers for vector embeddings, implementing cosine similarity for semantic retrieval and **FastAPI** for CRUD operations
- Architected a data pipeline using **Google Gemini** for image context extraction, optimizing latency by caching **vector embeddings** and image encoded data in MongoDB to eliminate redundant API data calls

### Data Science Club Website | *TypeScript, Next.js, React, PostgreSQL, Turborepo*

May 2025

- Built a monorepo using **Turborepo** with shared UI components, types, and configs to eliminate code duplication, centralize CI/CD, and securely isolate admin and member **Next.js** apps serving **500+** students
- Improved query efficiency by **40%** by migrating from **MongoDB** to **PostgreSQL** with normalized schemas, indexes, and **Row-Level Security**, refactoring backend to a layered **Controller-Service-Repository** architecture for modularity and maintainability

## SKILLS

- **Languages:** Python, C/C++, JavaScript/TypeScript, SQL, Java, Bash, HTML, CSS
- **Frameworks:** React, Next.js, Angular, Flask, FastAPI, Express.js, Tailwind CSS, Jest, Playwright
- **Tools:** AWS, PostgreSQL, MongoDB, Node.js, Docker, Git, Postman, Figma, GitHub Actions, Storybook